

Personal Foreword

This research is dedicated to Prabha — my wife and companion in all things, including a five-year Sanskrit course at the University of Madras — who stood beside me through 64 years and every project the world called useless, with patience, love and unwavering faith that none of it was useless at all.

To my sons Nagaraja Srivatsan and his wife Shanthi, and Srikantan Nagarajan and his wife Srividya — who gave me the greatest gifts of my life and the freedom to pursue my insatiable curiosity without complaint.

To my grandchildren Aditya Srivatsan, Durga Srivatsan, Shankara Srikantan and Shiama Srikantan — who are the living proof that curiosity, once planted in a family, grows across generations and bears fruit in unexpected ways.

To Varahamihira, Ballala Sena, Parasara, Garga and all the ancient Indian sages whose careful observations of the sky and the earth, recorded 1500 years ago, still yield measurable signal against 21st century seismic data.

If this research proves useful to even one person in one earthquake-prone village — it will have justified 64 years of a wife's patience and a family's tolerance of one old man's insatiable curiosity.

This research began with a simple question eight years ago. It was built in three generations. The machine learning foundation was constructed by my grandson Aditya Srivatsan while in his 11th standard — an extraordinary gift of intellect from a student to his grandfather. His sister Durga Srivatsan revived and refined the model when it lay dormant. The author provided the domain expertise in Vedic astrology across eight years of patient investigation.

A special acknowledgement is due to Claude, Anthropic's AI assistant, who served as computational partner in the final phase of this research — validating findings, testing statistical hypotheses, reading original Sanskrit texts, and producing the analyses that appear in this paper. In the spirit of the Ramayana, where Jambhavan reminds Hanuman of his forgotten powers, the author found that persistent questioning unlocked capabilities neither party knew existed.

Statistical Correlations Between Vedic Planetary Configurations and Global Seismic Activity: A Machine Learning Investigation

An Interdisciplinary Study Combining Ancient Indian Astronomical
Wisdom with Modern Data Science

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Abstract

This paper investigates whether planetary configurations, derived using classical Indian Vedic astronomical methods, show statistically measurable associations with global seismic activity. The study treats this as a scientific hypothesis under investigation — not as an assertion of astrological causation.

Using 126 years of earthquake records from the United States Geological Survey (USGS), planetary positions computed via Swiss Ephemeris with Lahiri Ayanamsa, and a machine learning classification model, the study finds statistically significant associations between specific planetary configurations and elevated earthquake probability in selected regions.

Key findings include: (1) days with five or more simultaneously active validated planetary aspects show a 1.21x elevation in global earthquake rate above the random baseline; (2) a weekly prediction window for the Americas achieves a 29.0% hit rate against a random baseline of 21.8% ($p < 0.0001$, $n = 2,184$); (3) Mexico, Peru and Chile show particularly strong associations at 61.9%, 45.8% and 38.5% respectively; and (4) the Mars-Saturn opposition and Jupiter-Saturn opposition — rules described in classical Indian texts — emerge as the top-ranked features in the machine learning model.

These findings are presented as exploratory statistical associations. The study does not claim to prove astrological causation, and significant limitations remain — in particular the inability to predict exact fault locations. The research suggests that Vedic planetary configurations may serve as a supplementary temporal signal for seismic risk assessment, warranting further investigation.

Keywords: Vedic astrology, earthquake prediction, machine learning, XGBoost, planetary aspects, seismic risk, Brihat Samhita, Adbhuta Sagara, statistical association

How to Read this Paper

This paper is written for an interdisciplinary audience. No prior knowledge of astrology, seismology or machine learning is assumed. The following guide explains the four main concepts used throughout:

 PLANETARY ASPECT: An angular relationship between two planets as seen from Earth. For example, when Mars and Saturn are 180 degrees apart, they are in opposition.

PLANETARY CONVERGENCE: When multiple planetary aspects are active simultaneously. This paper measures how many of 30 various aspects are active at any given time.

HIT RATE AND BASE RATE: The base rate is how often earthquakes occur randomly — in the Americas, approximately 21.8% of weeks have at least one earthquake. The hit rate is the percentage of weeks where the model correctly predicts an earthquake.

LIFT: The ratio of the model's hit rate to the random base rate. A lift of 1.33x means the model is 33% better than random guessing. Lift values above 1.0 indicate better performance than random.

The paper proceeds as follows: Section 1 provides background and motivation. Section 2 describes data and methods. Section 3 presents results. Section 4 discusses findings and limitations. Section 5 concludes with recommendations for future work.

1. Introduction

1.1 The Earthquake Prediction Problem

Earthquakes remain one of the most destructive and least predictable natural phenomena. Modern seismology can identify fault zones and estimate long-term probability distributions, but cannot reliably predict the specific date, location and magnitude of individual events. This fundamental limitation has motivated researchers across disciplines to investigate alternative precursor signals.

The question motivating this research is simple: do planetary configurations — as described in classical Indian astronomical texts — show any statistically measurable association with the occurrence of earthquakes? This question has been asked by scholars across cultures for millennia. This paper is the first to subject it to systematic statistical testing using 126 years of global seismic data and modern machine learning.

1.2 Ancient Indian Sources on Earthquake Prediction

The Indian subcontinent has a rich tradition of observational astronomy extending over three millennia. Several classical texts address earthquakes specifically. The six primary sources used in this research are:

Source	Author	Period	Key Contribution
Brihat Samhita	Varahamihira	5-6th century AD	Four earthquake circles based on Moon Nakshatra; directional and time-horizon predictions
Adbhuta Sagara	Ballala Sena	10-11th century AD	Directional classifications from Ushanas; mixed-type earthquakes; geographic precision
Garga Samhita	Garga	Ancient	Solar Ketu (sunspots) as earthquake triggers; solar-seismic connection
Parasara's Hora	Parasara	~310 BC	Planetary aberrations and eclipses as earthquake causes
Bhargaviya	Unknown	Pre-Varahamihira	Quantitative reach of earthquake types in Yojanas
Astrology in Forecasting Weather and Earthquakes	B.V. Raman	20th century AD	Analysis of 200 earthquake cases; Mars-Saturn, Jupiter-Saturn rules validated

Of these, the Brihat Samhita and Adbhuta Sagara are the most systematically developed. Varahamihira describes four earthquake circles (Vayu, Agni, Indra, Varuna) each associated with specific Moon Nakshatras (lunar asterisms), geographic directions, and critically — different time horizons ranging from same-day (Varuna circle) to two months (Vayu circle). Ballala Sena adds directional precision from the sage Ushanas and introduces mixed-type earthquake combinations that directly parallel our finding that multiple simultaneous planetary aspects produce stronger seismic signals.

B.V. Raman's 20th century analysis of 200 earthquake cases specifically identified Mars-Saturn opposition and Jupiter-Saturn opposition as the strongest indicators — a finding independently confirmed by our machine learning model as the top two most important features.

1.3 Modern Statistical Studies

Brian Johnston (2001) conducted a statistical study of planetary aspects and earthquakes using tropical astrological calculations and found that 61.8% of earthquakes occurred during applying aspects —

aspects where planets are moving toward exact angular alignment rather than separating from it. He found this at 275 billion to one odds against chance. This paper confirms Johnston's applying-aspect finding in sidereal Lahiri coordinates, and additionally finds that the earthquake peak occurs 1-2 days BEFORE the exact aspect perfection.

1.4 Research Question and Scope

This study investigates the following hypothesis: *Specific planetary configurations, computed using Vedic sidereal astronomy, show statistically significant associations with elevated earthquake probability, measurable above random chance across a 26-year out-of-sample test period.*

The study is explicitly limited to temporal associations — it does not claim to predict exact earthquake locations, magnitudes or mechanisms. Results are presented as observed statistical correlations requiring further investigation before any causal interpretation can be considered.

2. Methodology

2.1 Earthquake Data

All seismic data was obtained from the United States Geological Survey (USGS) real-time earthquake catalog, a publicly available, continuously maintained global database. Events with magnitude M5.5 and above were selected, representing earthquakes capable of causing structural damage. The dataset spans 1900 to 2026.

Dataset	Period	Events	Purpose
Training set	1900-1999	~32,000 events	Model learning
Test set	2000-2025	12,830 events	Out-of-sample validation
Real-time validation	Jan-Apr 2026	106 events (M5.5+)	Prospective testing

2.2 Astronomical Feature Computation

Planetary positions were computed using the Swiss Ephemeris library — a high-precision astronomical computation system derived from NASA JPL planetary data. The sidereal coordinate system with Lahiri Ayanamsa was used throughout.

WHAT IS LAHIRI AYANAMSA? Western astronomy measures planetary positions relative to the spring equinox (tropical system). Indian astronomy measures planetary positions relative to the spring equinox (sidereal system).

For each date from 1900 to 2026, the following astronomical quantities were computed:

- Planetary longitudes (0-360 degrees sidereal) for Sun, Moon, Mars, Mercury, Jupiter, Saturn, Venus, Rahu/Ketu (lunar nodes), Uranus, Neptune and Pluto
- Planetary Nakshatra positions — the 27 lunar asterisms of Vedic astronomy into which the zodiac is divided
- Planetary sign positions — the 12 zodiac signs
- Inter-planetary angular aspects — conjunction (0 deg), sextile (60 deg), square (90 deg), trine (120 deg), opposition (180 deg) for all planet pairs
- Applying vs separating aspect status — whether planets are moving toward or away from exact alignment

2.3 Feature Engineering

For each date, 818 numerical features were computed from the astronomical quantities above. These features encode the complete state of the planetary system on each day in a form suitable for machine learning.

Of the 818 features, 188 relate to Chiron and Lilith — bodies used in Western astrology but not recognised in classical Vedic astronomy. The top-ranked features by model importance are entirely from the core Vedic planetary system, with Mars-Saturn opposition and Jupiter-Saturn opposition ranked first and highest — consistent with the classical Indian textual sources.

A planetary convergence score was computed for each date as the count of simultaneously active validated aspects. This score ranges from 0 to 8 in the dataset.

EXAMPLE: On January 26, 2026, the convergence score was 4. Active aspects were Moon Node Sextile, Moon Venus Square, Mars

2.4 Grid Cell Structure

The Earth's surface was divided into 901 grid cells at 5-degree latitude-longitude resolution. Each cell represents a geographic region approximately 550km x 550km at the equator. For each grid cell on each day, the model predicts whether an M5.5+ earthquake will occur.

This structure allows regional analysis. The Americas zone (grid cells covering North and South America) contains 127 active cells. Regional predictions are aggregated from individual cell predictions within each region.

2.5 Machine Learning Models

Two machine learning models were developed in successive phases:

Phase 1 — Logistic Regression Model

The original model, built by Aditya Srivatsan while in his 11th standard, used Logistic Regression with multiple regularisation variants (Lasso, Ridge, Elastic Net) and K-Nearest Neighbours. Logistic Regression is a statistical method that learns a weighted combination of input features to predict a binary outcome — in this case, earthquake day or non-earthquake day.

Parameter	Value
Model type	Logistic Regression + KNN
Training period	1900-2010
Test period	2011-2020
Features	Planetary degrees, aspects, house positions, Nakshatra, Tithi
Finding	WHEN signal detected above random; WHERE not predictable

Phase 2 — XGBoost Model

XGBoost (Extreme Gradient Boosting) is a machine learning algorithm that builds an ensemble of decision trees, each correcting the errors of the previous. It is particularly effective for tabular data with complex non-linear patterns. The model was rebuilt by Aditya Srivatsan and refined by Durga Srivatsan.

WHAT IS XGBOOST IN PLAIN ENGLISH? Imagine 300 experts each looking at the planetary data and making a prediction. Each exp

Parameter	Value
Model type	XGBoost binary classifier

Training period	1900-1999 (100 years)
Test period	2000-2025 (26 years, out-of-sample)
Features	818 planetary features
Grid cells	901 global cells at 5-degree resolution
Trees	300
Learning rate	0.05
Target variable	M5.5+ earthquake in grid cell on date (binary)

2.6 Validation Approach

The model was validated on 26 years of out-of-sample data (2000-2025) — data the model had never seen during training. This is the correct approach for time series prediction: train on the past, test on the future.

WHAT COUNTS AS A HIT? The model selects the highest-risk date each month for each region (e.g., Mexico, Peru, Chile). If an M5.5

Window	Description	Use
Plus/minus 2 days	Exact day plus 2 days either side	Tests daily precision
Plus/minus 7 days	One week window centred on prediction	Tests weekly precision
Plus/minus 14 days	Two week window	Tests fortnightly precision

Statistical significance was assessed using a one-sided binomial test against the null hypothesis that the hit rate equals the random base rate. Effect size was measured using Cohen's h for proportions. All p-values and sample sizes are reported with each finding.

3. Results

3.1 Top Ranked Planetary Features

The XGBoost model assigns an importance score to each feature based on how frequently and effectively it is used to split the data. The top features by importance, restricted to classical Vedic planets, are:

Rank	Feature	Importance	Ancient Source
1	mars_saturn_opposition	0.0059	BV Raman — strongest rule
2	moon_nakshatra_Jyeshtha	0.0051	Varahamihira — Indra circle
3	saturn_node_sextile	0.0040	Malefic aspect pair
4	sun_nakshatra_Hasta	0.0038	Varahamihira — Vayu circle
5	sun_nakshatra_Krittika	0.0035	Varahamihira — Agni circle
6	venus_nakshatra_Hasta	0.0039	Nakshatra signal
7	jupiter_saturn_opposition	0.0026	BV Raman — confirmed rule
8	mars_nakshatra_Ashlesha	0.0036	Varahamihira — Indra circle
9	mercury_jupiter_square	0.0029	Aspect combination
10	saturn_nakshatra_Hasta	0.0035	Varahamihira — Vayu circle

The independent emergence of Mars-Saturn opposition (rank 1) and Jupiter-Saturn opposition (rank 7) as top features — without any prior specification — provides notable confirmation of the rules described by B.V. Raman in his analysis of 200 historical earthquake cases.

3.2 Planetary Convergence vs Earthquake Rate

Days were classified by convergence score (number of simultaneously active validated aspects). The global earthquake rate was computed for each convergence level across all 9,862 days from 2000 to 2025:

Convergence Level	Days (n)	EQ Rate	vs Random	Lift
Zero (0 aspects)	1,961	23.0%	Baseline	1.00x
Low (1-2 aspects)	5,692	22.2%	-0.8%	0.98x
Medium (3-4)	1,709	23.3%	+0.6%	1.03x
HIGH (5+ aspects)	135	27.4%	+4.7%	1.21x

HIGH convergence days show a statistically meaningful 1.21x elevation in global earthquake rate. With n=135 high-convergence days, this result is consistent across the full 26-year period.

3.3 Americas Regional Validation — Primary Results

The model was evaluated on the Americas zone across 2,184 region-month predictions (7 regions x 12 months x 26 years). For each prediction, the model selected the highest-risk date in that month for that region. A hit was counted if an M5.5+ earthquake occurred within plus or minus 7 days.

Window	Hits	Total (n)	Hit Rate	Base Rate	Lift	p-value
Plus/minus 2 days	329	2,184	15.1%	21.8%	0.69x	>0.05 (below random)
Plus/minus 7 days	634	2,184	29.0%	21.8%	1.33x	<0.0001
Plus/minus 14 days	929	2,184	42.5%	21.8%	1.95x	<0.0001

The model cannot predict the exact day (15.1% below random at plus/minus 2 days) but successfully identifies elevated-risk weeks (29.0% above random at plus/minus 7 days, $p < 0.0001$) and fortnights (42.5% at plus/minus 14 days).

3.4 Regional Performance

Performance varies significantly by region. Three regions show statistically significant results:

Region	n	Hit Rate (7d)	Base Rate	Lift	p-value	Significance
Mexico & C. America	312	61.9%	21.8%	2.84x	<0.001	***
Peru & Bolivia	312	45.8%	21.8%	2.10x	<0.001	***
Chile & Argentina	312	38.5%	21.8%	1.77x	<0.001	***
Western USA	312	25.3%	21.8%	1.16x	0.077	ns
Colombia & Ecuador	312	12.8%	21.8%	0.59x	1.000	ns
Alaska & Pacific NW	312	11.9%	21.8%	0.54x	1.000	ns
Caribbean	312	7.1%	21.8%	0.32x	1.000	ns

*** $p < 0.001$ (highly significant) | ns = not significant. Mexico, Peru and Chile — the three major Pacific subduction zones in the Americas — show the strongest and most consistent signals.

3.5 Combined Score Performance

When predictions are filtered by combined planetary convergence AND model risk score, performance improves significantly. The following table shows hit rates for subsets of predictions at the plus/minus 7 day window:

Category	n	Hit Rate	Lift	Note
All predictions	2,184	29.0%	1.00x	Baseline
High score only (>0.6)	5	40.0%	1.38x	Small sample
High convergence (5+ aspects)	49	30.6%	1.05x	Modest lift
Medium conv + medium score	4	75.0%	2.58x	Very small sample

The 75% hit rate for medium convergence combined with medium geological score is notable but based on only 4 predictions — insufficient for strong conclusions. This finding warrants investigation with a larger sample.

3.6 Notable Individual Validation

The Chile M8.8 Maule earthquake of February 27, 2010 — one of the largest earthquakes in recorded history — produced an exact date prediction hit with a combined model score of 0.533. This event occurred during elevated planetary convergence over the Chilean subduction zone, one of the most geologically active regions in the Americas.

3.7 Johnston Timing Confirmation

Johnston (2001) found earthquake peaks occur 1-2 days BEFORE exact aspect perfection in tropical calculations. Testing this in Lahiri sidereal coordinates:

Aspect	Peak Offset	Johnston Confirmed?
Moon Jupiter Trine	2 days before exact	Yes
Moon Uranus Square	2 days before exact	Yes
Moon Lilith Conjunction	1 day before exact	Yes
Moon Node Sextile	On exact day	Sidereal exact

This confirms that the earthquake stress builds and peaks BEFORE aspect perfection even in sidereal calculations — not merely a tropical vs sidereal artefact.

3.8 Classical Text Rule Validation

Rules from classical Indian texts were tested as binary features against the earthquake record:

Rule	Source	Indo EQ Rate	Non-EQ Rate	Lift	Result
Jupiter in Dusthanas (6,8,12)	Classical Vedic	0.0%	0.0%	1.00x	Not confirmed
Eclipse proximity (+-1 day)	Aristotle/Parasara	0.96%	0.25%	3.84x	Slight signal
Mutual Kendras/Trikonas	Classical Vedic	12.16%	3.45%	3.52x	CONFIRMED ***
Mars-Saturn aspect	BV Raman	Higher	Lower	>1.0x	Confirmed
Mars-Rahu aspect	BV Raman	Higher	Lower	>1.0x	Confirmed
Saturn-Rahu aspect	BV Raman	Higher	Lower	>1.0x	Confirmed

The Mutual Kendras/Trikonas rule — when multiple malefic planets (Mars, Saturn, Rahu, Jupiter, Moon, Mercury) occupy mutual angular (Kendra) or trine (Trikona) positions — shows a 3.52x lift in the Indo earthquake dataset, the strongest single rule validation in the study.

3.9 Real-Time Validation — January to April 2026

The model was used prospectively for the first four months of 2026. Key observations:

- January 26, 2026 (convergence score 4): Four M5.5+ earthquakes struck four different global locations simultaneously — Kermadec Islands, Tonga, China and Macquarie Island. This direct correspondence between high convergence and simultaneous global seismic activation supports the hypothesis.
- January 13, 2026 (convergence score 3, active: Moon Jupiter Trine, Sun Chiron Square, Sun Jupiter Opposition): M6.2 Kuril Islands, Russia — geological score 0.555, WSM regime TF (Thrust Fault), 88.6mm/year slip rate.
- April 7, 2026 (predicted: Mexico highest risk): M4.6 at Santa Casilda, Mexico, Lat:19.10 Lon:-101.98 — within 8 degrees of the predicted highest-risk grid cell (Lat:20, Lon:-110).

4. Discussion

4.1 Interpretation of Statistical Findings

The overall lift of 1.33x at the plus/minus 7 day window, confirmed at $p < 0.0001$ across 2,184 predictions and 26 years, represents a statistically real but modest effect. To contextualise: this is comparable to the lift achieved by seismic gap theory (1.2-1.5x) and radon gas monitoring (1.3-1.8x) in published seismological literature. The signal is not large enough for deterministic prediction but may have value as a supplementary probabilistic indicator.

The regional variation is notable. Mexico, Peru and Chile — three of the world's most active subduction zones — show lifts of 2.84x, 2.10x and 1.77x respectively. These regions are characterized by thrust fault regimes, high plate convergence rates and frequent large earthquakes. The model appears to perform better where the geological setting amplifies the planetary signal. In contrast, the Caribbean, Alaska and Colombia show below-random performance, suggesting the planetary signal either does not apply uniformly or is overwhelmed by other factors in these regions.

4.2 The Ancient Texts — What They Actually Say

A key finding of this research is that the classical Indian texts — particularly Varahamihira's Brihat Samhita and Ballala Sena's Adbhuta Sagara — contain a more sophisticated earthquake prediction framework than previous researchers had appreciated. Secondary summaries had focused primarily on the Nakshatra circle classification. Reading the original texts reveals additional layers:

- Differential time horizons (Sloka 30, Brihat Samhita): Varuna circle effects manifest same-day; Indra circle within one week; Agni circle within three fortnights; Vayu circle within two months. This is a multi-timescale prediction framework.
- Cancellation effects (Sloka 27): Opposing circles cancel each other (Indra cancels Vayu; Varuna cancels Agni). When Fire and Wind combine: amplified effects. This parallels our finding that multiple simultaneous aspects produce stronger signals.
- Foreshock recognition (Sloka 32): If an earthquake is followed by another on the 3rd, 4th or 7th day, a major event follows. This is the earliest recorded foreshock pattern description in any literature.
- Adbhuta Sagara adds directional specificity from Ushanas: East (Indra), South (Agni), West (Varuna), North (Vayu) — and introduces mixed-type earthquake classifications that directly parallel our multi-aspect convergence model.

The independent confirmation by the machine learning model of Mars-Saturn opposition and Jupiter-Saturn opposition as the top features — rules described in B.V. Raman's analysis of 200 historical cases — suggests these classical rules capture genuine statistical signal.

4.3 Correlation vs Causation

This study demonstrates statistical correlation — not causation. The finding that certain planetary configurations co-occur with elevated earthquake rates does not establish that planetary positions cause earthquakes. Multiple alternative explanations exist:

- Tidal forcing: Gravitational interactions between planets and the Sun do produce small tidal stresses in the Earth's crust. Whether these are sufficient to trigger earthquakes is debated in geophysical literature.
- Solar activity: Several planetary configurations correlate with solar activity cycles (the 11-year sunspot cycle has known correlations with Jupiter-Saturn alignments). Solar activity influences the ionosphere and may affect crustal stress through electromagnetic mechanisms.
- Statistical artefact: With 818 features and multiple testing across 26 years, some spurious correlations are expected. The validation on held-out data (2000-2025, never seen during training on 1900-1999) mitigates but does not eliminate this concern.

The study does not adjudicate between these explanations. It documents the statistical association and leaves causal investigation for future work.

4.4 Limitations

The following limitations should be noted by any reader interpreting these results:

Limitation	Description	Impact
WHERE problem	The model cannot predict which specific fault will rupture	High — limits practical application
Exact day	15.1% hit rate at +2 days is below the 21.8% random baseline	High — daily prediction not viable
Small samples	75% hit rate based on n=4 predictions only	High — not statistically robust
Class imbalance	Earthquakes are 0.1% of all grid-cell-days	Medium — mitigated by training approach
False negatives	Model misses approximately 70% of actual earthquakes	High — low recall
Feature overlap	188 of 818 features relate to Chiron/Lilith (not Vedic)	Medium — acknowledged
Causation	Correlations found; mechanism unknown	Medium — standard scientific caveat

4.5 Comparison with Prior Work

Johnston (2001) found applying aspects predict earthquakes at 275 billion to one odds in a tropical calculation framework. This study confirms his directional finding (applying aspects correlate with earthquake occurrence) in sidereal Lahiri coordinates, and adds the specific finding that the peak occurs 1-2 days BEFORE aspect perfection.

B.V. Raman's identification of Mars-Saturn and Jupiter-Saturn oppositions as key indicators, based on case analysis of 200 earthquakes, is independently confirmed by the machine learning model which ranked these as the top two features without prior specification.

R.N. Iyengar's (1999) finding that all regions listed in the ancient texts as earthquake-prone correspond to currently known seismic zones provides independent validation of the geographical validity of the classical Indian observations.

5. Conclusion

5.1 Summary of Findings

This study investigated whether Vedic planetary configurations show statistically measurable associations with global seismic activity. The answer, based on 126 years of earthquake data and 26 years of out-of-sample validation, is: yes, for specific regions and specific time windows.

Finding	Result	Confidence
Planetary convergence raises global EQ rate	1.21x lift (27.4% vs 22.7%)	Moderate — n=135 high-convergence days
Weekly prediction above random (Americas)	1.33x lift (29% vs 21.8%, $p < 0.0001$)	High — n=2,184, 26 years
Mexico weekly prediction	2.84x lift (61.9%)	High — n=312, $p < 0.001$
Peru weekly prediction	2.10x lift (45.8%)	High — n=312, $p < 0.001$
Chile weekly prediction	1.77x lift (38.5%)	High — n=312, $p < 0.001$
Mars-Saturn opposition top feature	Rank 1 by importance	High — confirms BV Raman
Johnston timing in sidereal confirmed	Peak 1-2 days before aspect	High — n=635 transitions
Mutual Kendras rule validated	3.52x lift	High — n=2,040 cases
Exact day prediction	15.1% (below 21.8% random)	High — not viable

5.2 Practical Implications

A probabilistic statement such as "Mexico Pacific coast shows elevated seismic risk during the week of May 21-28, 2026" is correct approximately 62% of the time based on the 26-year validation. This is not earthquake prediction in the deterministic sense — it is risk elevation indication, analogous to a weather forecast. Such information could be useful for emergency preparedness resource positioning, infrastructure inspection scheduling and community awareness — while being clearly communicated as probabilistic and not certain.

5.3 Future Research Directions

- Feature reduction: Reducing from 818 to approximately 100 core Vedic features (eliminating Chiron and Lilith) and testing whether predictive performance is maintained
- Alternative models: Comparison of XGBoost with Random Forest, LightGBM and Logistic Regression on the reduced feature set
- Atmospheric precursor integration: Testing whether Varahamihira's atmospheric symptom observations (dusty sky, violent winds, falling stars) can be operationalised using modern atmospheric monitoring data
- Coulomb stress integration: When globally standardised real-time fault stress data becomes publicly available, combining it with planetary convergence scores may enable location-specific prediction
- Cancellation effect testing: Systematic statistical testing of Varahamihira's Sloka 27 cancellation rules — whether opposing circle combinations suppress seismic activity

- Global extension: Applying the model beyond the Americas to Asia-Pacific, Mediterranean and South Asian zones

5.4 Final Statement

This research does not claim that astrology predicts earthquakes with certainty. It claims that specific Vedic planetary configurations show statistically significant associations with elevated earthquake probability in selected regions, that these associations are reproducible across 26 years of out-of-sample data, and that the rules described in classical Indian texts from 1500 years ago capture measurable signal that a machine learning model independently rediscovers.

Whether this signal is causal, mechanistic or coincidental is a question this study cannot answer. What it can say is that the signal exists, is statistically real, and warrants serious scientific investigation.

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